Hypersonic Interceptor Guidance, Thermal Dynamics, and Matrix Kinematics

Advanced Undergraduate Physics & Aerospace Engineering Seminar

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Problem Statement

A hypersonic guided interceptor vehicle with mass m and principal tensor of inertia $\mathbf{I}_b = \mathrm{diag}(I_{xx}, I_{yy}, I_{zz})$ in its body-fixed frame $\mathcal{B}$ operates within a non-isothermal, exponentially decaying atmosphere modeled by:

$$\rho(z) = \rho_0 \exp\left(-\frac{z}{H}\right), \quad T(z) = T_0 - \lambda z \tag{1}$$

where ρ_0 is sea-level air density, H is atmospheric scale height, T_0 is sea-level temperature, and λ is lapse rate.

The vehicle targets a maneuverable exo-/endo-atmospheric target using a **Pure Proportional Navigation (PPN)** guidance scheme in 3D Euclidean space $\mathbb{R}^3$.

Part A: 3D Kinematics, Lie Groups, and Quaternion Dynamics

1. **Kinematic Matrix Dynamics & Quaternion Invariance:** Let $\mathbf{q} = \begin{bmatrix} q_0 & q_1 & q_2 & q_3 \end{bmatrix}^T \in \mathbb{S}^3$ be the unit quaternion representing orientation mapping vectors from the inertial frame $\mathcal{I}$ to the body frame $\mathcal{B}$. The instantaneous angular velocity in body coordinates is $\boldsymbol{\omega}_b = \begin{bmatrix} \omega_x & \omega_y & \omega_z \end{bmatrix}^T$.

   (a) Derive the quaternion kinematic system in matrix-vector form:

   $$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega}(\boldsymbol{\omega}_b)\mathbf{q} \tag{2}$$

   explicitly constructing the 4×4 skew-symmetric matrix $\mathbf{\Omega}(\boldsymbol{\omega}_b)$.

   (b) Prove rigorously using multivariable calculus that the spatial norm constraint $\|\mathbf{q}(t)\|_2^2 = 1$ is invariant for all $t \geq 0$ under this dynamical operator by evaluating:

   $$\frac{d}{dt}\langle \mathbf{q}, \mathbf{q} \rangle = 2\mathbf{q}^T\dot{\mathbf{q}} \tag{3}$$

2. **Line-of-Sight (LOS) Kinematics & Vector Calculus:** Let $\mathbf{r}(t) = \mathbf{r}_t(t) - \mathbf{r}_v(t) \in \mathbb{R}^3$ denote the relative position vector between target and vehicle, $R(t) = \|\mathbf{r}(t)\|_2$ represent range, and $\mathbf{u}(t) = \frac{\mathbf{r}(t)}{R(t)} \in \mathbb{S}^2$ be the LOS unit vector.

   (a) Prove that the LOS angular velocity vector $\mathbf{\Omega}_{\text{LOS}}$ satisfies:

   $$\mathbf{\Omega}_{\text{LOS}} = \mathbf{u} \times \dot{\mathbf{u}} = \frac{\mathbf{r} \times \mathbf{v}_{\text{rel}}}{R^2} \tag{4}$$

   where $\mathbf{v}_{\text{rel}} = \dot{\mathbf{r}} = \mathbf{v}_t - \mathbf{v}_v$.

   (b) Derive the complete second time-derivative expression $\ddot{\mathbf{u}}$ as a function of R, $\dot{R}$, $\mathbf{\Omega}_{\text{LOS}}$, and relative acceleration $\mathbf{a}_{\text{rel}} = \ddot{\mathbf{r}}$.

Part B: Multivariable Aerodynamic Mechanics & Line-Integral Thermodynamics

3. Aerodynamic Force Transformation & Jacobian Evaluation: Aerodynamic forces generated in the velocity wind frame $\mathcal{W}$ are given by:

$$\mathbf{F}_w = \begin{bmatrix} -D \\ Y \\ -L \end{bmatrix} = \frac{1}{2}\rho(z)\|\mathbf{v}_v\|_2^2 S_{\text{ref}} \begin{bmatrix} -(C_{D0} + K\alpha^2 + K_\beta \beta^2) \\ C_{Y\beta}\beta \\ -(C_{L0} + C_{L\alpha}\alpha) \end{bmatrix} \tag{5}$$

where $\alpha = \arctan\left(\frac{v_{bz}}{v_{bx}}\right)$ and $\beta = \arcsin\left(\frac{v_{by}}{\|\mathbf{v}_v\|}\right)$.

(a) Compute the explicit Jacobian matrix mapping linear body velocity variations $\mathbf{v}_b = \begin{bmatrix} v_{bx} & v_{by} & v_{bz} \end{bmatrix}^T$ to aerodynamic angle variations:

$$\mathbf{J}_{\alpha,\beta} = \frac{\partial(\alpha, \beta)}{\partial \mathbf{v}_b} \in \mathbb{R}^{2\times 3} \tag{6}$$

evaluated at equilibrium symmetric trim flight $(\alpha_0, \beta_0 = 0)$.

4. Stagnation-Point Line-Integral Thermal Accumulation: Convective aerodynamic heat flux at the nose cap radius R_n follows the Fay-Riddell formulation:

$$q_s(t) = C_H \sqrt{\frac{\rho(z(t))}{R_n}} \|\mathbf{v}_v(t)\|_2^3 \tag{7}$$

(a) Consider a ballistic trajectory segment parameterized by $t \in [0, T_f]$ where $z(t) = z_0 - (v_0 \sin\gamma_0)t$ and velocity decays exponentially due to drag: $\|\mathbf{v}_v(t)\|_2 = v_0 e^{-kt}$. Evaluate the total integrated thermal energy absorbed per unit area $Q_{\text{total}} = \int_{\mathcal{C}} q_s(t)\, dt$ in closed form using definite integration:

$$Q_{\text{total}} = C_H \sqrt{\frac{\rho_0}{R_n}} v_0^3 \int_0^{T_f} \exp\left(\frac{v_0 \sin\gamma_0 t - z_0}{2H} - 3kt\right) dt \tag{8}$$

Part C: Linearized State-Space Stability & Eigenvalue Spectral Analysis

5. Coupled Longitudinal/Lateral State-Space Dynamics: The perturbed coupled pitch-yaw small-signal dynamics are represented in state-space form as $\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}_c(t)$ with state vector $\mathbf{x} = \begin{bmatrix}\Delta\alpha & \Delta q & \Delta\beta & \Delta r\end{bmatrix}^T \in \mathbb{R}^4$:

$$\mathbf{A} = \begin{bmatrix} -\frac{Z_\alpha}{V_0} & 1 & 0 & 0 \\ M_\alpha & M_q & 0 & 0 \\ 0 & 0 & \frac{Y_\beta}{V_0} & -1 \\ 0 & 0 & N_\beta & N_r \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\frac{Z_\delta}{V_0} & 0 \\ M_\delta & 0 \\ 0 & \frac{Y_\delta}{V_0} \\ 0 & N_\delta \end{bmatrix} \tag{9}$$

(a) Utilizing block matrix determinant identities:

$$\det \begin{bmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22} \end{bmatrix} = \det(\mathbf{A}_{11}) \det(\mathbf{A}_{22}) \tag{10}$$

compute the 4th-order characteristic polynomial $P(\lambda) = \det(\lambda\mathbf{I} - \mathbf{A}) = 0$.

(b) Derive necessary and sufficient conditions on stability derivatives $(Z_\alpha, M_\alpha, M_q)$ using the **Routh-Hurwitz Criterion** applied to the pitch block $\mathbf{A}_{11}$ to guarantee strictly negative real parts for all longitudinal eigenvalues $(\mathrm{Re}(\lambda_i) < 0)$.

(c) Formulate the pole-placement algebraic equation to calculate full-state feedback matrix $\mathbf{K} \in \mathbb{R}^{2\times4}$ under control law $\mathbf{u}_c = -\mathbf{K}\mathbf{x}$ to assign the closed-loop spectrum $\sigma(\mathbf{A} - \mathbf{B}\mathbf{K}) = \{\mu_1, \mu_2, \mu_3, \mu_4\} \subset \mathbb{C}^-$.

Part D: Strategic Mindmap & Mental Framework

Meta-Strategy: Conceptual Deconstruction

When confronting a high-dimensional, multi-physics challenge combining kinematics, vector calculus, aerothermodynamics, and linear control theory, **never start writing equations right away**. Divide the challenge into three distinct cognitive passes:

I. **Kinematic Phase (Part A):** Focus strictly on frames of reference, rotations, and differential geometry of trajectories. Aerodynamics and forces do not exist yet.

II. **Dynamic & Thermodynamic Phase (Part B):** Introduce physical fields (atmosphere, velocity fields) and compute differential mappings (Jacobians) and cumulative energetic impacts (Line Integrals).

III. **Systems Phase (Part C):** Freeze non-linearities, linearize around equilibrium, and evaluate spectral stability (eigenvalues and matrix structures).

1. Mindmap for Part A: Kinematics & Vector Mechanics

Problem 1: Quaternion Kinematics & Identity Proofs

→ **Mental Picture:** Imagine a 3D body rotating continuously in space. A quaternion $\mathbf{q}$ tracks this orientation without coordinate singularities (gimbal lock). The angular velocity vector $\boldsymbol{\omega}_b$ acts as a generator of rotation.

→ **Calculus Tool:** Matrix product rule and dynamic system conservation laws.

→ **Approach Strategy:**

1. Recall that quaternion multiplication $\mathbf{q} \otimes \boldsymbol{\omega}$ can be rewritten as a matrix-vector multiplication $\boldsymbol{\Omega}(\boldsymbol{\omega}_b)\mathbf{q}$.
2. Expand the cross-product components of rotational kinematics into an explicit 4×4 skew-symmetric matrix.
3. To prove normative invariance $\|\mathbf{q}(t)\|_2 = 1$, compute $\frac{d}{dt}(\mathbf{q}^T\mathbf{q}) = 2\mathbf{q}^T\dot{\mathbf{q}}$. Substitute $\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}\mathbf{q}$ and exploit the fundamental matrix property that $\mathbf{x}^T\mathbf{M}\mathbf{x} = 0$ for any skew-symmetric matrix $\mathbf{M}$.

Problem 2: Line-of-Sight (LOS) Kinematics & Relative Geometry

→ **Mental Picture:** Two moving points (Interceptor and Target) connected by a dynamic line segment of changing length $R(t)$ and changing pointing angle $\mathbf{u}(t)$.

→ **Vector Calculus Tool:** Product rule applied to vector norms and cross products.

→ **Approach Strategy:**

1. Express relative position as $\mathbf{r} = R\mathbf{u}$. Take the first derivative with respect to time using the product rule: $\mathbf{v}_{\text{rel}} = \dot{R}\mathbf{u} + R\dot{\mathbf{u}}$.
2. Take the cross product of $\mathbf{u}$ with $\mathbf{v}_{\text{rel}}$ to eliminate the scalar range-rate term $\dot{R}\mathbf{u}$ (since $\mathbf{u} \times \mathbf{u} = \mathbf{0}$).
3. Recognize that the instantaneous rotation rate vector of a unit vector obeys $\boldsymbol{\Omega}_{\text{LOS}} = \mathbf{u} \times \dot{\mathbf{u}}$.
4. For $\ddot{\mathbf{u}}$, differentiate $\dot{\mathbf{u}} = \frac{\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u}}{R}$ systematically using the quotient rule in vector form.

2. Mindmap for Part B: Aerodynamics & Line Integration

Problem 3: Multivariable Dynamic Mapping (Jacobians)

→ **Mental Picture:** The vehicle's velocity components in the body frame (v_{bx}, v_{by}, v_{bz}) dictate the aerodynamic incident angles α (attack) and β (sideslip). We need the local linear sensitivity matrix around trimmed equilibrium.

→ **Multivariable Calculus Tool:** Multi-variable partial differentiation and Matrix Jacobians.

→ **Approach Strategy:**

1. Set up the vector field mapping $\mathbf{g} : \mathbb{R}^3 \to \mathbb{R}^2$ where $\mathbf{g}(\mathbf{v}_b) = \begin{bmatrix} \alpha(\mathbf{v}_b) \\ \beta(\mathbf{v}_b) \end{bmatrix}^T$.
2. Apply partial derivative rules for inverse trigonometric functions:
$$\frac{d}{dx}(\arctan(x)) = \frac{1}{1+x^2}, \quad \frac{d}{dx}(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}}$$
3. Chain-rule each term with respect to v_{bx}, v_{by}, and v_{bz}.
4. Evaluate the final matrix at symmetric flight conditions where $v_{by} = 0 \implies \beta_0 = 0$.

Problem 4: Path Integrals of Exponential Fields

→ **Mental Picture:** A vehicle diving through a density-stratified atmosphere while decelerating. The thermal load is a non-linear scalar field along a 1D curve in 4D space-time.

→ **Calculus 2 Tool:** Parameterized definite line integrals and exponential integrations.

→ **Approach Strategy:**

1. Express altitude $z(t)$ and velocity $V(t)$ explicitly as scalar functions of time t.
2. Substitute $z(t)$ into the density function $\rho(z) = \rho_0 e^{-z/H}$, taking special care of the square-root term $\sqrt{\rho(z)} = \sqrt{\rho_0} e^{-z/(2H)}$.
3. Substitute $V(t) = v_0 e^{-kt}$ raised to the 3rd power: $V(t)^3 = v_0^3 e^{-3kt}$.
4. Combine all exponential exponents into a single unified exponent of the form e^{At-B}.
5. Factor out all constants and execute a standard single-variable integration $\int e^{At} dt = \frac{1}{A} e^{At}$ evaluated from $t = 0$ to $t = T_f$.

3. Mindmap for Part C: Linear Systems & Stability

Problem 5: Block Matrices, Characteristic Polynomials, & Control Synthesis

→ **Mental Picture:** High-dimensional coupled differential equations decomposed into independent structural subsystems (Pitch plane vs. Yaw plane).

→ **Linear Algebra & Control Tools:** Block determinant identities, Routh-Hurwitz criteria, and pole-placement matrix algebra.

→ **Approach Strategy:**

1. **Block Determinants:** Observe that matrix $\mathbf{A}$ is block-diagonal with zero cross-coupling terms between longitudinal $(\Delta\alpha, \Delta q)$ and lateral $(\Delta\beta, \Delta r)$ axes. Use $\det(\mathbf{A}) = \det(\mathbf{A}_{11})\det(\mathbf{A}_{22})$ to split a 4th-order polynomial calculation into two simpler 2nd-order problems.
2. **Routh-Hurwitz Stability:** For a quadratic sub-polynomial $\lambda^2 + a_1\lambda + a_0 = 0$, asymptotic stability requires all coefficients to be strictly positive $(a_1 > 0, a_0 > 0)$. Translate these algebraic conditions back into the physical aerodynamic stability derivatives $(Z_\alpha, M_\alpha, M_q)$.
3. **Pole Placement Synthesis:** To design the feedback gains in $\mathbf{u}_c = -\mathbf{K}\mathbf{x}$:
   - Compute the desired target characteristic equation from roots $\prod_{i=1}^{4}(\lambda - \mu_i) = 0$.
   - Form the symbolic matrix $(\mathbf{A} - \mathbf{B}\mathbf{K})$ and compute its characteristic polynomial $\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K}))$.
   - Equate corresponding coefficients to form a solvable system of linear algebraic equations for elements of matrix $\mathbf{K}$.

Part E: Complete Step-by-Step Mathematical Solutions & Detailed Explanations

Problem 1: Quaternion Kinematics & Norm Invariance Proof

(a) Derivation of $\dot{\mathbf{q}} = \frac{1}{2}\boldsymbol{\Omega}(\boldsymbol{\omega}_b)\mathbf{q}$

A unit quaternion representing an orientation mapping from the inertial frame $\mathcal{I}$ to the body frame $\mathcal{B}$ is defined as:

$$\mathbf{q} = \begin{bmatrix} q_0 \\ \mathbf{q}_v \end{bmatrix} = \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \cos(\theta/2) \\ \hat{\mathbf{n}}\sin(\theta/2) \end{bmatrix} \tag{11}$$

where θ is the rotation angle and $\hat{\mathbf{n}} = [n_x, n_y, n_z]^T$ is the unit vector along the axis of rotation.

The time rate of change of orientation under a angular velocity vector $\boldsymbol{\omega}_b = [\omega_x, \omega_y, \omega_z]^T$ in the body-fixed frame is given by the fundamental quaternion differential equation:

$$\dot{\mathbf{q}} = \frac{1}{2}\mathbf{q} \otimes \bar{\boldsymbol{\omega}}_b \tag{12}$$

where $\bar{\boldsymbol{\omega}}_b = [0, \omega_x, \omega_y, \omega_z]^T$ is the pure quaternion formed from the angular velocity, and $\otimes$ denotes quaternion multiplication.

Quaternion Multiplication Rule: For two quaternions $\mathbf{p} = [p_0, \mathbf{p}_v]^T$ and $\mathbf{s} = [s_0, \mathbf{s}_v]^T$:

$$\mathbf{p} \otimes \mathbf{s} = \begin{bmatrix} p_0 s_0 - \mathbf{p}_v \cdot \mathbf{s}_v \\ p_0\mathbf{s}_v + s_0\mathbf{p}_v + \mathbf{p}_v \times \mathbf{s}_v \end{bmatrix} \tag{13}$$

Setting $\mathbf{p} = \mathbf{q} = [q_0, q_1, q_2, q_3]^T$ and $\mathbf{s} = \bar{\boldsymbol{\omega}}_b = [0, \omega_x, \omega_y, \omega_z]^T$:

1. **Scalar Part Calculation:**

$$\text{Scalar} = q_0(0) - (q_1\omega_x + q_2\omega_y + q_3\omega_z) = -\omega_x q_1 - \omega_y q_2 - \omega_z q_3 \tag{14}$$

2. **Vector Part Calculation:**

$$\text{Vector} = q_0 \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} + 0 \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} + \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix} \times \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \tag{15}$$

Evaluating the vector cross product explicitly:

$$\mathbf{q}_v \times \boldsymbol{\omega}_b = \begin{bmatrix} q_2\omega_z - q_3\omega_y \\ q_3\omega_x - q_1\omega_z \\ q_1\omega_y - q_2\omega_x \end{bmatrix} \tag{16}$$

Summing the components for each vector entry:

$$\dot{q}_1 = \frac{1}{2}(q_0\omega_x + q_2\omega_z - q_3\omega_y) \tag{17}$$

$$\dot{q}_2 = \frac{1}{2}(q_0\omega_y + q_3\omega_x - q_1\omega_z) \tag{18}$$

$$\dot{q}_3 = \frac{1}{2}(q_0\omega_z + q_1\omega_y - q_2\omega_x) \tag{19}$$

Combining scalar and vector components into matrix form $\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega}(\boldsymbol{\omega}_b)\mathbf{q}$:

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} \tag{20}$$

Thus, the explicit 4×4 skew-symmetric kinematic matrix is:

$$\mathbf{\Omega}(\boldsymbol{\omega}_b) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \tag{21}$$

(b) Proof of Unit Norm Invariance $\|\mathbf{q}(t)\|_2^2 = 1$

Let $S(t) = \|\mathbf{q}(t)\|_2^2 = \mathbf{q}^T\mathbf{q} = q_0^2 + q_1^2 + q_2^2 + q_3^2$. We evaluate the time derivative using the matrix product rule:

$$\frac{d}{dt}\left(\mathbf{q}^T\mathbf{q}\right) = \dot{\mathbf{q}}^T\mathbf{q} + \mathbf{q}^T\dot{\mathbf{q}} \tag{22}$$

Substitute $\dot{\mathbf{q}} = \frac{1}{2}\mathbf{\Omega}\mathbf{q}$ and $\dot{\mathbf{q}}^T = \left(\frac{1}{2}\mathbf{\Omega}\mathbf{q}\right)^T = \frac{1}{2}\mathbf{q}^T\mathbf{\Omega}^T$:

$$\frac{d}{dt}\left(\mathbf{q}^T\mathbf{q}\right) = \frac{1}{2}\mathbf{q}^T\mathbf{\Omega}^T\mathbf{q} + \frac{1}{2}\mathbf{q}^T\mathbf{\Omega}\mathbf{q} = \frac{1}{2}\mathbf{q}^T\left(\mathbf{\Omega}^T + \mathbf{\Omega}\right)\mathbf{q} \tag{23}$$

By inspecting the matrix $\mathbf{\Omega}(\boldsymbol{\omega}_b)$, we check skew-symmetry ($\mathbf{\Omega}^T = -\mathbf{\Omega}$):

$$\mathbf{\Omega}^T = \begin{bmatrix} 0 & \omega_x & \omega_y & \omega_z \\ -\omega_x & 0 & -\omega_z & \omega_y \\ -\omega_y & \omega_z & 0 & -\omega_x \\ -\omega_z & -\omega_y & \omega_x & 0 \end{bmatrix} = -\mathbf{\Omega} \tag{24}$$

Therefore:

$$\mathbf{\Omega}^T + \mathbf{\Omega} = -\mathbf{\Omega} + \mathbf{\Omega} = \mathbf{0}_{4\times4} \tag{25}$$

Substituting this back into the derivative:

$$\frac{d}{dt}\|\mathbf{q}(t)\|_2^2 = \frac{1}{2}\mathbf{q}^T\left(\mathbf{0}_{4\times4}\right)\mathbf{q} = 0 \tag{26}$$

Since $\frac{d}{dt}\|\mathbf{q}(t)\|_2^2 = 0$, the norm $\|\mathbf{q}(t)\|_2^2$ is constant for all t. Given that $\mathbf{q}(0)$ is initialized as a unit quaternion ($\|\mathbf{q}(0)\|_2 = 1$), $\|\mathbf{q}(t)\|_2 = 1$ is invariant for all time. $\blacksquare$

Problem 2: Line-of-Sight (LOS) Kinematics Vector Derivations

(a) Proof of $\mathbf{\Omega}_{\mathbf{LOS}} = \mathbf{u} \times \dot{\mathbf{u}} = \frac{\mathbf{r}\times\mathbf{v}_{\mathbf{rel}}}{R^2}$

By definition, the relative position vector $\mathbf{r}$ is expressed in terms of range $R = \|\mathbf{r}\|_2$ and unit directional vector $\mathbf{u} \in \mathbb{S}^2$:

$$\mathbf{r} = R\mathbf{u} \implies \mathbf{u} = \frac{\mathbf{r}}{R} \tag{27}$$

Differentiating $\mathbf{r} = R\mathbf{u}$ with respect to time using the vector product rule:

$$\mathbf{v}_{\text{rel}} = \dot{\mathbf{r}} = \frac{d}{dt}(R\mathbf{u}) = \dot{R}\mathbf{u} + R\dot{\mathbf{u}} \tag{28}$$

To isolate $\dot{\mathbf{u}}$, cross-multiply the entire equation from the left by $\mathbf{u}$:

$$\mathbf{u} \times \mathbf{v}_{\text{rel}} = \mathbf{u} \times (\dot{R}\mathbf{u} + R\dot{\mathbf{u}}) = \dot{R}(\mathbf{u} \times \mathbf{u}) + R(\mathbf{u} \times \dot{\mathbf{u}}) \tag{29}$$

Since any vector crossed with itself is zero ($\mathbf{u} \times \mathbf{u} = \mathbf{0}$):

$$\mathbf{u} \times \mathbf{v}_{\text{rel}} = R(\mathbf{u} \times \dot{\mathbf{u}}) \implies \mathbf{u} \times \dot{\mathbf{u}} = \frac{\mathbf{u} \times \mathbf{v}_{\text{rel}}}{R} \tag{30}$$

Now substitute $\mathbf{u} = \frac{\mathbf{r}}{R}$ into the right-hand side:

$$\mathbf{u} \times \dot{\mathbf{u}} = \frac{\left(\frac{\mathbf{r}}{R}\right) \times \mathbf{v}_{\text{rel}}}{R} = \frac{\mathbf{r} \times \mathbf{v}_{\text{rel}}}{R^2} \tag{31}$$

By definition, the angular velocity vector of a unit vector's orientation frame is given by $\mathbf{\Omega}_{\text{LOS}} = \mathbf{u} \times \dot{\mathbf{u}}$, proving:

$$\mathbf{\Omega}_{\text{LOS}} = \mathbf{u} \times \dot{\mathbf{u}} = \frac{\mathbf{r} \times \mathbf{v}_{\text{rel}}}{R^2} \quad \blacksquare \tag{32}$$

(b) Derivation of Second Derivative $\ddot{\mathbf{u}}$

Rearranging $\mathbf{v}_{\text{rel}} = \dot{R}\mathbf{u} + R\dot{\mathbf{u}}$ to solve for $\dot{\mathbf{u}}$:

$$\dot{\mathbf{u}} = \frac{\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u}}{R} = \frac{1}{R}\mathbf{v}_{\text{rel}} - \frac{\dot{R}}{R}\mathbf{u} \tag{33}$$

Differentiating $\dot{\mathbf{u}}$ with respect to time using quotient and product rules:

$$\ddot{\mathbf{u}} = \frac{d}{dt}\left(\frac{\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u}}{R}\right) = \frac{R\frac{d}{dt}(\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u}) - (\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u})\dot{R}}{R^2} \tag{34}$$

Evaluating the numerator derivative term $\frac{d}{dt}(\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u})$:

$$\frac{d}{dt}(\mathbf{v}_{\text{rel}} - \dot{R}\mathbf{u}) = \mathbf{a}_{\text{rel}} - (\ddot{R}\mathbf{u} + \dot{R}\dot{\mathbf{u}}) \tag{35}$$

Substitute this back into the expression for $\ddot{\mathbf{u}}$:

$$\ddot{\mathbf{u}} = \frac{R(\mathbf{a}_{\text{rel}} - \ddot{R}\mathbf{u} - \dot{R}\dot{\mathbf{u}}) - \dot{R}\mathbf{v}_{\text{rel}} + \dot{R}^2\mathbf{u}}{R^2} \tag{36}$$

$$= \frac{\mathbf{a}_{\text{rel}} - \ddot{R}\mathbf{u} - 2\dot{R}\dot{\mathbf{u}}}{R} \tag{37}$$

(using $\mathbf{v}_{\text{rel}} = \dot{R}\mathbf{u} + R\dot{\mathbf{u}}$ to combine terms).

Expressing $\dot{\mathbf{u}}$ in terms of $\mathbf{\Omega}_{\text{LOS}}$: since $\mathbf{u}\cdot\mathbf{u} = 1 \implies \mathbf{u}\cdot\dot{\mathbf{u}} = 0$, cross-multiplying $\mathbf{\Omega}_{\text{LOS}} = \mathbf{u}\times\dot{\mathbf{u}}$ by $\mathbf{u}$ yields:

$$\mathbf{\Omega}_{\text{LOS}} \times \mathbf{u} = (\mathbf{u} \times \dot{\mathbf{u}}) \times \mathbf{u} = (\mathbf{u} \cdot \mathbf{u})\dot{\mathbf{u}} - (\dot{\mathbf{u}} \cdot \mathbf{u})\mathbf{u} = \dot{\mathbf{u}} \tag{38}$$

Substituting $\dot{\mathbf{u}} = \mathbf{\Omega}_{\text{LOS}} \times \mathbf{u}$ into the acceleration vector yields the complete expanded form:

$$\ddot{\mathbf{u}} = \frac{\mathbf{a}_{\text{rel}} - \ddot{R}\mathbf{u} - 2\dot{R}(\mathbf{\Omega}_{\text{LOS}} \times \mathbf{u})}{R} \tag{39}$$

Problem 3: Multivariable Dynamic Mapping & Jacobian Computation

The aerodynamic angles of attack α and sideslip β are defined by velocity components in the body frame $\mathbf{v}_b = [v_{bx}, v_{by}, v_{bz}]^T$:

$$\alpha(\mathbf{v}_b) = \arctan\left(\frac{v_{bz}}{v_{bx}}\right) \tag{40}$$

$$\beta(\mathbf{v}_b) = \arcsin\left(\frac{v_{by}}{\|\mathbf{v}_v\|}\right) \quad \text{where } \|\mathbf{v}_v\| = \sqrt{v_{bx}^2 + v_{by}^2 + v_{bz}^2} \tag{41}$$

We compute the 2×3 Jacobian matrix $\mathbf{J}_{\alpha,\beta} = \begin{bmatrix} \frac{\partial \alpha}{\partial v_{bx}} & \frac{\partial \alpha}{\partial v_{by}} & \frac{\partial \alpha}{\partial v_{bz}} \\ \frac{\partial \beta}{\partial v_{bx}} & \frac{\partial \beta}{\partial v_{by}} & \frac{\partial \beta}{\partial v_{bz}} \end{bmatrix}$.

1. Partial Derivatives of Angle of Attack α

Using $\frac{d}{dx}(\arctan(x)) = \frac{1}{1+x^2}$:

1. **With respect to v_{bx}:**

$$\frac{\partial \alpha}{\partial v_{bx}} = \frac{1}{1 + \left(\frac{v_{bz}}{v_{bx}}\right)^2} \cdot \left(-\frac{v_{bz}}{v_{bx}^2}\right) = \frac{v_{bx}^2}{v_{bx}^2 + v_{bz}^2} \cdot \left(-\frac{v_{bz}}{v_{bx}^2}\right) = -\frac{v_{bz}}{v_{bx}^2 + v_{bz}^2} \tag{42}$$

2. **With respect to v_{by}:**

$$\frac{\partial \alpha}{\partial v_{by}} = 0 \quad \text{(since } \alpha \text{ has no dependence on } v_{by}\text{)} \tag{43}$$

3. **With respect to v_{bz}:**

$$\frac{\partial \alpha}{\partial v_{bz}} = \frac{1}{1 + \left(\frac{v_{bz}}{v_{bx}}\right)^2} \cdot \left(\frac{1}{v_{bx}}\right) = \frac{v_{bx}^2}{v_{bx}^2 + v_{bz}^2} \cdot \frac{1}{v_{bx}} = \frac{v_{bx}}{v_{bx}^2 + v_{bz}^2} \tag{44}$$

2. Partial Derivatives of Sideslip Angle β

Using $\frac{d}{dx}(\arcsin(x)) = \frac{1}{\sqrt{1-x^2}}$ and letting $f = \frac{v_{by}}{V}$ where $V = \sqrt{v_{bx}^2 + v_{by}^2 + v_{bz}^2}$:

$$\frac{\partial \beta}{\partial v_i} = \frac{1}{\sqrt{1 - \left(\frac{v_{by}}{V}\right)^2}} \cdot \frac{\partial}{\partial v_i}\left(\frac{v_{by}}{V}\right) = \frac{V}{\sqrt{V^2 - v_{by}^2}} \cdot \frac{\partial}{\partial v_i}\left(\frac{v_{by}}{V}\right) \tag{45}$$

At symmetric trim flight, $\beta_0 = 0 \implies v_{by} = 0$. Evaluating $\frac{V}{\sqrt{V^2 - v_{by}^2}}$ at $v_{by} = 0$:

$$\left.\frac{V}{\sqrt{V^2 - 0}}\right|_{v_{by}=0} = 1 \tag{46}$$

Now evaluating the partial derivatives of $f = \frac{v_{by}}{V}$ at $v_{by} = 0$:

1. **With respect to v_{bx}:**

$$\left.\frac{\partial}{\partial v_{bx}}\left(\frac{v_{by}}{V}\right)\right|_{v_{by}=0} = \left.-\frac{v_{by}}{V^2}\frac{\partial V}{\partial v_{bx}}\right|_{v_{by}=0} = 0 \tag{47}$$

2. **With respect to v_{by}:**

$$\frac{\partial}{\partial v_{by}}\left(\frac{v_{by}}{V}\right) = \frac{V(1) - v_{by}\frac{v_{by}}{V}}{V^2} = \frac{V^2 - v_{by}^2}{V^3} \xrightarrow{v_{by}=0} \frac{V^2}{V^3} = \frac{1}{V_0} \tag{48}$$

3. **With respect to v_{bz}:**

$$\frac{\partial}{\partial v_{bz}}\left(\frac{v_{by}}{V}\right)\bigg|_{v_{by}=0} = -\frac{v_{by}}{V^2}\frac{\partial V}{\partial v_{bz}}\bigg|_{v_{by}=0} = 0 \tag{49}$$

3. Assembling the Jacobian Matrix at Trim

Noting that $v_{bx} = V_0 \cos\alpha_0$ and $v_{bz} = V_0 \sin\alpha_0$ when $v_{by} = 0$, we simplify the first row:

$$v_{bx}^2 + v_{bz}^2 = V_0^2 \cos^2\alpha_0 + V_0^2 \sin^2\alpha_0 = V_0^2 \tag{50}$$

$$\frac{\partial\alpha}{\partial v_{bx}} = -\frac{V_0 \sin\alpha_0}{V_0^2} = -\frac{\sin\alpha_0}{V_0} \tag{51}$$

$$\frac{\partial\alpha}{\partial v_{bz}} = \frac{V_0 \cos\alpha_0}{V_0^2} = \frac{\cos\alpha_0}{V_0} \tag{52}$$

Assembling the complete linearized Jacobian matrix evaluated at trim:

$$\mathbf{J}_{\alpha,\beta}\Big|_{trim} = \begin{bmatrix} -\frac{\sin\alpha_0}{V_0} & 0 & \frac{\cos\alpha_0}{V_0} \\ 0 & \frac{1}{V_0} & 0 \end{bmatrix} \tag{53}$$

Problem 4: Closed-Form Line Integration of Heat Accumulation

The instantaneous heat flux is given by Fay-Riddell equation:

$$q_s(t) = C_H \sqrt{\frac{\rho(z(t))}{R_n}} \|\mathbf{v}_v(t)\|_2^3 \tag{54}$$

Given the state trajectories:

$$z(t) = z_0 - (v_0 \sin\gamma_0)t \tag{55}$$

$$\|\mathbf{v}_v(t)\|_2 = v_0 e^{-kt} \tag{56}$$

$$\rho(z) = \rho_0 e^{-z/H} \tag{57}$$

1. Expressing the Integrand in terms of t

First, compute $\sqrt{\rho(z(t))}$:

$$\sqrt{\rho(z(t))} = \sqrt{\rho_0 \exp\left(-\frac{z_0 - (v_0 \sin\gamma_0)t}{H}\right)} = \sqrt{\rho_0}\exp\left(-\frac{z_0 - (v_0 \sin\gamma_0)t}{2H}\right) \tag{58}$$

Next, compute $\|\mathbf{v}_v(t)\|_2^3$:

$$\|\mathbf{v}_v(t)\|_2^3 = (v_0 e^{-kt})^3 = v_0^3 e^{-3kt} \tag{59}$$

Substitute both expressions into $q_s(t)$:

$$q_s(t) = C_H \sqrt{\frac{\rho_0}{R_n}} \exp\left(-\frac{z_0}{2H} + \frac{v_0 \sin\gamma_0}{2H} t\right) v_0^3 \exp(-3kt) \tag{60}$$

Combine constants and combine exponents using $e^A e^B = e^{A+B}$:

$$q_s(t) = C_H v_0^3 \sqrt{\frac{\rho_0}{R_n}} \exp\left(-\frac{z_0}{2H}\right) \exp\left[\left(\frac{v_0 \sin\gamma_0}{2H} - 3k\right) t\right] \tag{61}$$

2. Evaluating the Definite Integral

Let $K_{\text{const}} = C_H v_0^3 \sqrt{\frac{\rho_0}{R_n}} e^{-\frac{z_0}{2H}}$ and $A = \frac{v_0 \sin\gamma_0}{2H} - 3k$. The total heat energy per unit area is:

$$Q_{\text{total}} = \int_0^{T_f} q_s(t)\,dt = K_{\text{const}} \int_0^{T_f} e^{At}\,dt \tag{62}$$

Using standard calculus integration $\int e^{At} dt = \frac{1}{A} e^{At}$:

$$Q_{\text{total}} = K_{\text{const}} \left[\frac{e^{At}}{A} \right]_0^{T_f} = \frac{K_{\text{const}}}{A} \left(e^{AT_f} - 1 \right) \tag{63}$$

Substituting back K_{const} and A:

$$Q_{\text{total}} = \frac{C_H v_0^3 \sqrt{\frac{\rho_0}{R_n}} \exp\left(-\frac{z_0}{2H}\right)}{\frac{v_0 \sin\gamma_0}{2H} - 3k} \left[\exp\left(\left[\frac{v_0 \sin\gamma_0}{2H} - 3k \right] T_f \right) - 1 \right] \tag{64}$$

Problem 5: Linear System Characteristic Analysis & Routh-Hurwitz Proof

(a) Characteristic Polynomial via Block Determinant

Given the system matrix $\mathbf{A} \in \mathbb{R}^{4\times 4}$:

$$\mathbf{A} = \begin{bmatrix} -\frac{Z_\alpha}{V_0} & 1 & 0 & 0 \\ M_\alpha & M_q & 0 & 0 \\ 0 & 0 & \frac{Y_\beta}{V_0} & -1 \\ 0 & 0 & N_\beta & N_r \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & \mathbf{A}_{22} \end{bmatrix} \tag{65}$$

The characteristic polynomial is defined as $P(\lambda) = \det(\lambda \mathbf{I}_4 - \mathbf{A})$:

$$\lambda \mathbf{I}_4 - \mathbf{A} = \begin{bmatrix} \lambda \mathbf{I}_2 - \mathbf{A}_{11} & \mathbf{0}_{2\times 2} \\ \mathbf{0}_{2\times 2} & \lambda \mathbf{I}_2 - \mathbf{A}_{22} \end{bmatrix} \tag{66}$$

Using the block determinant theorem $\det \begin{bmatrix} \mathbf{X} & \mathbf{0} \\ \mathbf{0} & \mathbf{Y} \end{bmatrix} = \det(\mathbf{X}) \det(\mathbf{Y})$:

$$P(\lambda) = \det(\lambda \mathbf{I}_2 - \mathbf{A}_{11}) \cdot \det(\lambda \mathbf{I}_2 - \mathbf{A}_{22}) = P_{\text{pitch}}(\lambda) \cdot P_{\text{yaw}}(\lambda) \tag{67}$$

Evaluating the pitch block determinant $P_{\text{pitch}}(\lambda)$:

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}_{11}) = \det \begin{bmatrix} \lambda + \frac{Z_\alpha}{V_0} & -1 \\ -M_\alpha & \lambda - M_q \end{bmatrix} \tag{68}$$

$$= \left(\lambda + \frac{Z_\alpha}{V_0} \right) (\lambda - M_q) - (-1)(-M_\alpha) \tag{69}$$

$$= \lambda^2 + \left(\frac{Z_\alpha}{V_0} - M_q \right) \lambda + \left(-\frac{Z_\alpha M_q}{V_0} - M_\alpha \right) \tag{70}$$

Similarly, evaluating the yaw block determinant $P_{\text{yaw}}(\lambda)$:

$$\det(\lambda \mathbf{I}_2 - \mathbf{A}_{22}) = \det \begin{bmatrix} \lambda - \frac{Y_\beta}{V_0} & 1 \\ -N_\beta & \lambda - N_r \end{bmatrix} \tag{71}$$

$$= \lambda^2 - \left(\frac{Y_\beta}{V_0} + N_r \right) \lambda + \left(\frac{Y_\beta N_r}{V_0} + N_\beta \right) \tag{72}$$

Multiplying both sub-polynomials gives the full 4th-order polynomial:

$$P(\lambda) = \left[\lambda^2 + \left(\frac{Z_\alpha}{V_0} - M_q \right) \lambda - \left(\frac{Z_\alpha M_q}{V_0} + M_\alpha \right) \right] \cdot \left[\lambda^2 - \left(\frac{Y_\beta}{V_0} + N_r \right) \lambda + \left(\frac{Y_\beta N_r}{V_0} + N_\beta \right) \right] \tag{73}$$

(b) Stability Analysis using Routh-Hurwitz Criterion

For a second-order polynomial $P(\lambda) = \lambda^2 + a_1\lambda + a_0 = 0$, the Routh array is constructed as follows: where $b_1 = \frac{a_1 \cdot a_0 - 1 \cdot 0}{a_1} = a_0$.

$$\begin{array}{c|cc} \lambda^2 & 1 & a_0 \\ \lambda^1 & a_1 & 0 \\ \lambda^0 & b_1 & 0 \end{array}$$

By the Routh-Hurwitz Theorem, all roots have strictly negative real parts ($\mathrm{Re}(\lambda_i) < 0$) if and only if all coefficients in the first column of the Routh array are strictly positive ($1 > 0$, $a_1 > 0$, $a_0 > 0$).

For the longitudinal/pitch block $P_{\text{pitch}}(\lambda) = \lambda^2 + a_1\lambda + a_0$:

$$a_1 = \frac{Z_\alpha}{V_0} - M_q \tag{74}$$

$$a_0 = -\frac{Z_\alpha M_q}{V_0} - M_\alpha \tag{75}$$

Applying the conditions:

1. **First Condition ($a_1 > 0$):**

$$\frac{Z_\alpha}{V_0} - M_q > 0 \implies M_q < \frac{Z_\alpha}{V_0} \tag{76}$$

   Since lift-curve slope gives $Z_\alpha > 0$ and pitch damping gives $M_q < 0$, this condition is physically satisfied by standard damped airframes.

2. **Second Condition ($a_0 > 0$):**

$$-\frac{Z_\alpha M_q}{V_0} - M_\alpha > 0 \implies M_\alpha < -\frac{Z_\alpha M_q}{V_0} \tag{77}$$

   Since $Z_\alpha > 0$ and $M_q < 0$, the term $-\frac{Z_\alpha M_q}{V_0}$ is positive. For static pitch stability, the pitching moment derivative with angle of attack M_α must be sufficiently negative ($M_\alpha < 0$, restoring moment).

Thus, the precise mathematical stability criteria are:

$$M_q < \frac{Z_\alpha}{V_0} \quad \text{and} \quad M_\alpha < -\frac{Z_\alpha M_q}{V_0} \quad \blacksquare \tag{78}$$

(c) Full State-Feedback Pole-Placement Formulation

Let the state-feedback law be $\mathbf{u}_c = -\mathbf{K}\mathbf{x}$, where $\mathbf{K} \in \mathbb{R}^{2\times 4}$ is defined as:

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & 0 & 0 \\ 0 & 0 & k_{23} & k_{24} \end{bmatrix} \tag{79}$$

The closed-loop system dynamics become $\dot{\mathbf{x}} = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}$.

Computing the closed-loop state matrix $\mathbf{A}_{cl} = \mathbf{A} - \mathbf{B}\mathbf{K}$:

$$\mathbf{B}\mathbf{K} = \begin{bmatrix} -\frac{Z_\delta}{V_0} & 0 \\ M_\delta & 0 \\ 0 & \frac{Y_\delta}{V_0} \\ 0 & N_\delta \end{bmatrix} \begin{bmatrix} k_{11} & k_{12} & 0 & 0 \\ 0 & 0 & k_{23} & k_{24} \end{bmatrix} = \begin{bmatrix} -\frac{Z_\delta k_{11}}{V_0} & -\frac{Z_\delta k_{12}}{V_0} & 0 & 0 \\ M_\delta k_{11} & M_\delta k_{12} & 0 & 0 \\ 0 & 0 & \frac{Y_\delta k_{23}}{V_0} & \frac{Y_\delta k_{24}}{V_0} \\ 0 & 0 & N_\delta k_{23} & N_\delta k_{24} \end{bmatrix} \tag{80}$$

Subtracting $\mathbf{BK}$ from $\mathbf{A}$ yields closed-loop block matrices $\mathbf{A}_{cl,11}$ and $\mathbf{A}_{cl,22}$:

$$\mathbf{A}_{cl,11} = \begin{bmatrix} -\frac{Z_\alpha - Z_\delta k_{11}}{V_0} & 1 + \frac{Z_\delta k_{12}}{V_0} \\ M_\alpha - M_\delta k_{11} & M_q - M_\delta k_{12} \end{bmatrix} \tag{81}$$

The closed-loop pitch characteristic equation is:

$$\det(\lambda \mathbf{I} - \mathbf{A}_{cl,11}) = \lambda^2 + \bar{a}_1(k_{11}, k_{12})\lambda + \bar{a}_0(k_{11}, k_{12}) \tag{82}$$

where:

$$\bar{a}_1 = \frac{Z_\alpha - Z_\delta k_{11}}{V_0} - (M_q - M_\delta k_{12}) \tag{83}$$

$$\bar{a}_0 = \left(\frac{Z_\alpha - Z_\delta k_{11}}{V_0}\right)(M_q - M_\delta k_{12}) - (1 + \frac{Z_\delta k_{12}}{V_0})(M_\alpha - M_\delta k_{11}) \tag{84}$$

Given target longitudinal poles $\mu_1, \mu_2 \in \mathbb{C}^-$, the desired target characteristic polynomial is:

$$P_{\text{target}}(\lambda) = (\lambda - \mu_1)(\lambda - \mu_2) = \lambda^2 - (\mu_1 + \mu_2)\lambda + \mu_1\mu_2 \tag{85}$$

Equating coefficients of $P_{\text{target}}(\lambda)$ and $\det(\lambda \mathbf{I} - \mathbf{A}_{cl,11})$ gives a system of two linear algebraic equations in terms of gains k_{11} and k_{12}:

$$\bar{a}_1(k_{11}, k_{12}) = -(\mu_1 + \mu_2) \tag{86}$$

$$\bar{a}_0(k_{11}, k_{12}) = \mu_1\mu_2 \tag{87}$$

Solving this linear matrix system (2×2) yields the unique gains k_{11} and k_{12}. Applying the identical algebraic procedure to the yaw block $\mathbf{A}_{cl,22}$ with target poles μ_3, μ_4 uniquely determines k_{23} and k_{24}.